

ASP3231 Project - The distance and age of Example

Person Name, Named Person and Name Surname
[Group number here]

[Optional pretty picture that doesn't need caption]

[No table of contents needed for <3500 words]

[This example report illustrates a structure that could work for your project report, but some variation is allowed. Do not copy-paste and slightly modify sentences directly from this report as that is effectively plagiarism.]

Abstract

We determine the distance and age of Example, a thingamy located in the constellation Whatsit. To do this we use photometry measured from J, H and K-band observations that we obtained with the THAT telescope on 1 April 2023 and 4 May 2023. To determine the distance of Example, we use the Fully-Tisher relation for luminosity as a function of colour, finding a distance of 360 kpc. Using measurements of the diameter of Example, we estimate it has an age of 92 ± 5 kyr, based on three diameter-age relations from the literature. Example is an example of a relatively nearby, young and low luminosity thingamy, but the methods used here can be used on older and higher luminosity things to determine redshift independent distances to 10 Mpc or more, making them particularly useful for cosmology.

Introduction

[Some material, particularly from hundreds of years ago, is now common knowledge and doesn't require referencing. For example, it is rare to provide a reference for Kepler's laws.]

Thingamies were first identified by Messiest in the late 18th Century, and are now known to be expanding clouds of gas caused by events. While the exact cause of the events is not known, thingamies are known to follow several scaling relations with relatively little scatter (e.g. Fully & Tisher 1967, Fubin & Rord 1975). The luminosity-colour relation of Fully & Tisher (1967) is of particular use for cosmology, as it allows estimation on thingamy luminosities with colours, with H-band luminosities having a 1-sigma scatter of just 20% as a function of J-H colour (e.g. Nould & Breeman 2004). Consequently, using large ensembles of thingamy distances, Nould & Breeman (2004) was able to determine the Hubble constant in the nearby Universe to be 74 km/s/Mpc.

Example was one of the first thingamies to be identified, being recorded in Carolynne Smerschel's catalogue of southern celestial objects in the early 19th century. Example is a

relatively young low luminosity thingamy, with an absolute magnitude of $M_V \sim -12$ (e.g. Pun 1996). However, as it is just 368 +/- 5 kpc from Earth it has a corresponding apparent magnitude of $V \sim 10$, making it detectable by relatively small telescopes and the subject of detailed studies by larger telescopes (e.g. Pun 1996).

In this work we determine the distance and age of Example using J, H and K-band images obtained with the This Hypothetical Astronomical Telescope (THAT) in 2023. Below we discuss the data, image processing, our measurements of the images and the physical properties of Example. We discuss Example and our methods in the context of other thingamies, before concluding.

Data and Data Processing

To measure the properties of Example we obtained J, H and K-band images with the THAT on 1 April 2023 and 4 May 2023. THAT is a 0.3141592 metre diameter telescope located at the Parkville Campus of Monash University. As there were clouds on both nights we obtained observations, we used flatfield exposures from 18 April 2023. Bias and dark exposures were obtained on all nights we obtained observations, as the bias and dark images from the THAT can vary from week-to-week (THAT Observing Guide).

[Paragraph or two describing the basic data processing, using words such as add, subtract, multiply and divide to convey understanding.]

[Paragraph or two describing the image shifting, scaling and combining. Beware of long paragraphs - a paragraph should typically have one key focus, topic or conclusion - very long paragraphs may be unfocused or contain too many topics.]

[Below is an example discussion of additional non-standard processing.]

As the bias and dark images from the THAT telescope can vary from week-to-week (THAT Instruction Guide) we visually inspected the processed images to look for evidence of poor bias and dark subtraction, including bright edges and hot pixels. For images from 1 April 2023, we did find that hot pixels were not adequately subtracted from the data. To correct for this, via a process of trial and error, we found doing dark subtraction with a median combined dark image multiplicatively scaled by 1.04 produced the dark subtraction for our science images.

Example is located 20 arcminutes from the bright star Ziriis, resulting in scattered light that produces a gradient across the background in our science images. To mitigate the impact of this on photometry, we have subtracted off the scattered light by fitting the background with a 2-dimensional polynomial (using astropy's Polynomial 2D). The final combined image for the H-band is shown in Figure 1.

[Paragraph or two on source catalogues and photometry, including calibration]

[H-band image]

Figure 1: The combined THAT telescope H-band image of the thingamy Example.

Distance

To determine the distance to Example we use the Fully & Tisher (1967) colour-luminosity relation. JHK apparent magnitudes and colours were determined using $r=200$ aperture photometry, which is sufficiently large to enclose the entire thingamy. While this does include several faint stars, the brightest of these stars is ~ 5 magnitudes fainter than Example and unlikely to impact the final results. The other properties of Thingamy are listed in Table 1.

To determine the distance to Example, we use the Fully-Tisher calibration of Nould & Breeman (2004) where:

$$M_H = -14.42 + 2.0 (J-H).$$

As we find $J-H=0.48$ the corresponding absolute magnitude of Example is $M_H = -13.46$. Using this apparent magnitude and the observed absolute magnitude of $H=9.32$, then using

$$M - M_H = 5 \log(d/10)$$

we find a corresponding distance (d) of 360 kpc.

Example is relatively bright and we expect photon shot noise to produce photometric uncertainties below 1%, so it is likely our errors will be dominated by systematic rather than random errors. The photometric calibration for our imaging was determined using 2MASS point source catalogue stars in our field, and the standard deviation of this calibration is approximately 0.02 mags in each band. To illustrate how this could impact our results, if we use $J=9.82$ and $H=9.30$ as the input apparent magnitudes, then $M_H = -13.38$ and the corresponding distance is 346 kpc. If the calibration is our primary source of error then our distance estimate and uncertainty are 360 ± 14 kpc.

Table 1: Observed properties of Example

Band	Magnitude	Radius (pixels)	Radius (Arcsec)
J	9.80	-	-
H	9.32	175	87
K	8.86	-	-

Radius

The age of thingamies is found to be tightly correlated with their observed radius, which is not unexpected given their expansion after the initial event. While the nature of the event is unknown, it seems to launch gas at an almost constant velocity into the intergalactic medium so age is almost directly proportional to radius. From the literature, the widely used age-radius relations are:

$$\log(t) = 1.12 \log(R) + 5.87$$

Alpher et al. (2007)

$$\log(t) = 1.16 \log(R) + 5.89$$

Bethe et al. (2009)

$$\log(t) = 1.08 \log(R) + 5.87$$

Gamov et al. (2014)

where t is in years and R is in kpc.

The first step towards determining the radius of Example is to identify its edges. In the literature a radius corresponding to an H-band surface brightness of 20 mag per square arcsecond is often used (e.g. Fould et al. 2004 and references therein). (While some works have used J and K-bands to define the radius, the bulk of the literature uses H-band so we have limited ourselves to this here.) To measure this boundary we generated a segmentation map where the signal-to-noise threshold corresponded to a $H=21.5$ (i.e. a flux 4 times fainter than $H=20.0$ given each pixel is 0.25 square arcseconds in area) and the minimum area was 9 pixels.

Using the segmentation map, we identified the x-y coordinates of pixels associated with Example (pixel value 10 in our segmentation map) that were neighbouring sky pixels (pixel value 0). We then fitted an ellipse to this data in python, and used the semi-major axis of the ellipse to define the radius. The x-y coordinates of the edges of Example and fitted ellipse are shown in Figure 2, and our final estimate of the radius is 175 pixels (87 arcseconds).

[Image showing fit of ellipse to data]

Figure 2: *The edge of Example defined using an H-band surface brightness of 20 mag per square arcsecond. Dots show the edge of Example defined using the segmentation map while the solid curve shows the elliptical fit.*

For our best estimate of Example's distance (360 kpc), we find that the corresponding radius is 0.152 kpc. As highlighted by Punnett (1996), this is far smaller than the typical thingamy which is over a kpc in radius. For the age-radius relations of Alpher et al. (2007), Bethe et al. (2009) and Gamov et al. (2014) the corresponding Example ages are 90 kyr, 87 kyr and 97 kyr. If limited by calibrations of the age-radius relation, our estimate of the age is 92 ± 5 kyr.

Discussion and Extension

We have observed Example and measured its photometry and angular size, and then used these measurements and thingamy relationships from the literature to find that Example has a distance of 360 ± 17 kpc and age of $92 \pm$ kyr. While our estimated age for Example is in good agreement with the literature (e.g. 368 ± 5 kpc, Punnett et al. 1996), our measurement of the age is only $\sim 75\%$ of common literature values (e.g. 135 ± 20 kyr, Fould et al. 2004 and references therein).

There are a number of reasons why our estimate of the age of Example is far lower than literature estimates. While we have measured ages with the widely used Alpher et al. (2007), Bethe et al. (2009) and Gamov et al. (2014) relations, these have been calibrated using mostly larger and older thingamies with radii greater than 1 kpc and ages more than a million years.

Environment may also be impacting our age estimate of Example. Example is a thingamy located within the Local Group of galaxies and may be in a relatively dense environment (e.g. Punnett et al. 1996). If Example's expansion is being slowed by dense gas, its size may be relatively small for its age, and this would explain our age underestimate. Spectroscopy can be used to mitigate these complications, as it can measure the current rate of thingamy expansion (using redshift and blueshift) and can provide some constraints on gas density (e.g. Punnett et al. 1996). However, as this project only used JHK imaging that was beyond the scope of our work.

Our distance estimate to Example, 360 ± 17 kpc, is very close to established distances from the literature (e.g. 368 ± 5 kpc, Punnett et al. 1996). This highlights that the Fully-Tisher relation for thingamies is relatively easy to use once one has measurements of the total apparent magnitudes and colours that are accurate to within a few hundredths of a magnitude.

Redshift independent distances for cosmology has been a major driver of JHK surveys of large swathes of sky such as the 3 Micron All Sky Survey (3MASS; Juchra et al. 2001). While redshift independent distances are useful for measuring the Hubble expansion of the Universe, they can also be used to measure the motions of galaxies towards concentrations of mass (bulk flows) such as galaxy clusters, which constrains the mass density of the Universe and how the mass density varies on large scales (~ 8 Mpc). Figure 3 illustrates the distribution of galaxies in the nearby Universe as traced by Cepheid, red giant branch and thingamy distances.

[Image showing distribution of galaxies]

Figure 3: The distribution of galaxies in the nearby Universe measured with the 3MASS survey and redshift independent distances, including those derived from thingamies. Mass concentrations are evident, particularly at $RA = 270^\circ$ and 8 Mpc. The redshift independent distances are combined with galaxy redshifts, bulk flows towards mass concentrations can be measured, which can constrain the mass density and distribution in the Universe.

Thingamies are particularly useful for cosmology as they are quite luminous - the bluest and brightest thingamies have $J-H \sim 0$ and $M_H \sim -14.4$, so if $H \sim 20$ thingamies can be detected and measured (i.e. with 3MASS) then their distances can be determined to 10 Mpc. There is the prospect of detecting thingamies to even greater distances with the advent of wide-field infrared space telescopes, which have a greatly reduced sky background compared to ground-based telescopes.

While properties of thingamies measured from imaging surveys are great tools for cosmology, there are some reasons for caution. As our own work highlights, the ages of thingamies can be grossly overestimated if they are in environments that are denser than average, such as the Local Group of galaxies and the Virgo Cluster of galaxies. As thingamy colours and luminosities are a function of age, this could result in environmentally dependent errors in thingamy distances. This may be mitigated by only using thingamies in a limited range of extragalactic environments or by using spectroscopy to directly measure the rates of thingamy expansion and better constrain ages (e.g. Punnett et al. 1996).

Conclusions

We have obtained and processed observations of the thingamy Example to determine its colour, absolute magnitude, distance, radius and age. We find that Example has an apparent magnitude and colour of $H=9.32$ and $J-H=0.48$, with a corresponding absolute magnitude of $M_H = -13.46$ and distance 360 ± 17 kpc (in good agreement with the recent literature).

We have modelled the projected edge of Example using an ellipse, and then used the corresponding radius and literature age-radius relations to determine Example's age. Our age estimate of 92 ± 5 kyr is well below literature values of 135 kyr, and this may be due to the expansion of Example being slowed by the relatively dense intergalactic medium in the Local Group of galaxies.

The methods we have used to measure the distance, radius and age of Example can be applied to more distant thingamies, enabling measurement of the Hubble constant and the density of the nearby Universe. However, the dependence of thingamy properties on environment (as illustrated by our age underestimate for Example) does highlight that some caution is required, and that spectroscopic observations of thingamies and modelling thingamy properties with environment is advisable.

Acknowledgements

This work made use of Jupyter, astropy, scipy and numpy software. We made use of code examples provided in the ASP3231 resources and Stackoverflow. We thank the TAs G Cop and B cop for their assistance.

References

[Use a reference style similar to astronomical journals, which can include abbreviations for journal names and generally doesn't include article titles. Some example references are provided below. The references should be alphabetical by author name whereas when you cite articles in your text you use chronological order - (e.g., Nould & Breeman 2004, Bethe 2009). When you cite an article with more than 3 authors use the first author name and et al., and you can use this in the references too if there's large numbers of authors.]

Alpher, X. 2007, ApJ, 123, 456

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This Hypothetical Astronomical Telescope (THAT) Observing Guide, 2022.

Appendix

This should only contain material that is useful but not essential to your final report. A reader shouldn't need to read the appendix to understand your report, but may refer to the appendix for future information.

[J, H and K images of Example and a figure caption]